

SUMMARY

Applied Machine Learning Engineer with 3+ years of experience building production-grade ML pipelines, with growing expertise in LLM fine-tuning and post-training workflows. Skilled in designing reinforcement learning environments, reward modeling, and evaluation frameworks for agentic applications. Experienced with deep learning architectures (LSTM, GRU, Transformer-based models), PyTorch, and end-to-end model training pipelines. Proven ability to balance research rigor with engineering execution from ablation studies and benchmark design to scalable deployment via REST APIs and cloud platforms (AWS, GCP, Databricks).

PROFESSIONAL EXPERIENCE

George Washington University

Washington, DC

ML/AI Engineer (IEEE Conference Paper – Under Review)

Aug 2025 - Dec 2025

- Architected an end-to-end forecasting pipeline on 177K+ real-world records, transforming 100+ unstructured HTML reports into structured, model-ready datasets demonstrating the full post-training data ingestion lifecycle at scale.
- Designed and executed systematic ablation studies comparing baseline and advanced model families (Linear Regression, XGBoost, FNN, GRU, LSTM), with a 4-layer LSTM (256 hidden units) achieving best performance (RMSE: 197→137, 30.6% improvement, $R^2 \approx 0.86$).
- Built reproducible training pipelines in PyTorch with logging, early stopping, dropout regularization, and modular architecture, enabling scalable experimentation directly analogous to post-training pipeline engineering for LLMs.
- Developed model evaluation metrics and benchmarks (RMSE, MAE, R^2) to accurately measure capability improvements across 100+ school-meal time series, reinforcing rigorous evaluation practices applicable to LLM alignment quality assessment.

George Washington University Law School

Washington, DC

Machine Learning Engineer (Student Technical Assistant)

July 2024 - Dec 2025

- Designed predictive models and evaluation signals to forecast classroom recording demand from historical scheduling and usage logs, improving resource allocation efficiency by 20%.
- Built classification models with reward-style signal design to predict high-risk recording conflicts (overlapping bookings, system overload), reducing last-minute failures by 25%.
- Developed automated anomaly detection logic with precision-recall trade-off analysis to identify abnormal access patterns, reducing manual troubleshooting by 30%.
- Implemented modular, reusable ML pipelines with experiment tracking across linear, tree-based, and neural network model families, supporting iterative model capability improvement.
- Deployed lightweight inference endpoints for real-time scheduling validation, integrating ML models into production operational systems.

Tata Consultancy Services

Bangalore, India

Machine Learning Engineer

Nov 2021 - Dec 2023

- Designed and implemented an anomaly detection framework on high-volume transactional data (1M+ records/day), reducing operational downtime by 30% demonstrating ability to scale and optimize distributed ML workloads.
- Engineered temporal and behavioral features (rolling counts, deviation from historical baselines, frequency spikes) to improve model separability, reflecting the feature engineering discipline applied in LLM post-training data pipelines.
- Trained and evaluated Isolation Forest and gradient-boosted models; calibrated thresholds using precision-recall trade-off analysis, reducing false positives by 25% while maintaining high detection sensitivity.
- Built reproducible training workflows with version-controlled experiments, consistent preprocessing, and logging practices directly transferable to fine-tuning and RL training run management.
- Collaborated with backend engineering teams to integrate ML models into production systems, ensuring reliable deployment, monitoring, and compatibility.

EDUCATION

George Washington University, Washington, DC

Jan 2024 - Dec 2025

Master of Science, Data Science

Coursework: Machine Learning, Deep Learning, NLP, Data Mining, Big Data Analytics, Cloud Computing

Dayananda Sagar University, Bangalore, India

Jun 2018 - Sep 2021

Bachelor's of Computer Applications

Coursework: Python, SQL, Databases, Programming, C, C++, Statistics, Data Analysis, Algorithms

SKILLS

Programming: Python, SQL, C/C++

LLMs & Post-Training: LLM Fine-Tuning, Supervised Fine-Tuning (SFT), Reinforcement Learning from Human Feedback (RLHF), Reward Modeling, Post-Training Pipelines, Transformer Architectures, Embeddings, Tokenization, RAG

RL & Evaluation: RL Environment Design, Evaluation Signal Development, Benchmark Design, Ablation Studies, Model Capability Assessment, Alignment Evaluation

Core ML: Supervised & Unsupervised Learning, Time Series Forecasting, Anomaly Detection, Feature Engineering, Hyperparameter Tuning, Regularization, Overfitting/Underfitting

Deep Learning: LSTM, GRU, FNN, Transformer-based NLP, Neural Architecture Design

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, MLflow, Git

MLOps & Deployment: Flask, REST APIs, Docker, AWS (S3, SageMaker), Model Serving, Distributed Training, Monitoring, Logging

Responsible AI: Fairness, Bias Mitigation, Data Privacy, Alignment Quality